Diabetes Prediction: A Comprehensive Study Integrating Deep Learning and Machine Learning Approaches

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Abstract

Diabetes, a prevalent chronic disease, affects a significant portion of the global population, with early detection and management being crucial for patient health. Despite the lack of a cure, advancements in machine learning (ML) and deep learning (DL) offer promising avenues for diabetes prediction. This study leverages the Pima Indian dataset to explore and compare the effectiveness of various ML and DL techniques, including Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), Random Forest, K-Nearest Neighbors (KNN), and Decision Tree. Our comprehensive evaluation, based on accuracy, precision, and recall, identified MLP as the most effective algorithm, achieving an accuracy rate of 85%. This research addresses current challenges in diabetes prediction by highlighting the superior performance of MLP in early detection, thereby underscoring its potential in improving diabetes management. The findings contribute to the ongoing discourse in healthcare technology, advocating for the integration of advanced ML techniques to enhance predictive accuracy and patient outcomes.

CCS Concepts

• **Computing methodologies → Machine learning approaches**; **Artificial intelligence**.

Keywords

Diabetes prediction, machine learning, Pima Indian Diabetic dataset, classification, Random Forest, SVM, Decision Tree, MLP, KNN, Healthcare.

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1 Introduction

Diabetes, a chronic metabolic disorder characterized by elevated blood sugar levels, poses a significant global health challenge [13]. Its prevalence continues to surge, with the World Health Organization (WHO) highlighting two primary types: type 1 and type 2 diabetes. Type 1 diabetes, often diagnosed in childhood, results from the body's immune system attacking insulin-producing cells, necessitating lifelong insulin therapy. In contrast, type 2 diabetes, which accounts for the majority of cases, typically develops in adults and is often associated with obesity and sedentary lifestyles, underscoring the importance of understanding its risk factors and mechanisms [3]. As the International Diabetes Federation projects, the current population of over 285 million individuals affected by diabetes is poised to escalate to a staggering 380 million within the next two decades [11]. This trajectory emphasizes the urgency of developing robust strategies for early detection and effective management. Given the intricate interplay of genetic, lifestyle, and environmental factors contributing to diabetes, predictive models hold promise in identifying high-risk individuals and implementing timely interventions to mitigate the disease's progression.

Early diabetes detection is essential because it allows for the prompt application of treatment and preventative measures, which eventually improves health outcomes, lowers healthcare costs, and improves quality of life for those with the disease [12]. By identifying diabetes in its early stages, healthcare providers can initiate lifestyle modifications and pharmacological treatments to mitigate the risk of complications, prevent hospitalizations, and promote overall well-being. Moreover, early diagnosis reduces the burden on caregivers and healthcare systems, fostering a more sustainable approach to diabetes management while empowering individuals with the knowledge and resources to manage their condition and lead fulfilling lives effectively [2]. In recent years, integrating advanced computational techniques, particularly deep learning and machine learning, has emerged as a promising avenue for enhancing the accuracy and reliability of diabetes diagnosis predictions. Deep learning and machine learning are revolutionizing diabetes detection by leveraging vast amounts of medical data to identify patterns and risk factors with high accuracy. Machine learning algorithms analyze features such as blood glucose levels, BMI, age, and genetic markers to predict the likelihood of diabetes onset. Deep learning, with its neural network architecture, can handle complex and unstructured data, including medical images and electronic health records, to detect subtle signs of the disease that may be overlooked by traditional methods. These advanced techniques enhance early diagnosis, personalized treatment plans, and overall patient outcomes by providing a more comprehensive and precise analysis of diabetes risk and presence. This capability positions deep learning and machine learning as indispensable tools for more precise and personalized approaches to diabetes management and prevention.

The dataset under scrutiny in this investigation is the Pima Indian dataset, a renowned resource frequently employed in diabetes research due to its decadent array of health-related parameters. Housed within the machine learning lab at the University of California, Irvine, this database is instrumental in assessing the effectiveness of data mining techniques for diabetes classification [8]. Researchers leverage its extensive data points and delve into the intricate interplay of variables contributing to diabetes onset. This study adopts a dual-pronged approach, integrating machine learning and deep learning methodologies to enhance diabetes prediction accuracy. Employing techniques such as Principal Component Analysis (PCA) [6], the research aims to distil critical features essential for effective machine learning models. Support Vector Machine (SVM), Random Forest, and Decision Tree models are trained to utilize the insights gleaned from PCA, while the K-Nearest Neighbors (KNN) algorithm undergoes optimization through grid search [5]. In tandem, a deep learning framework is implemented using a Multi-Layer Perceptron (MLP) architecture. The efficacy of these models is meticulously assessed across multiple performance metrics, including Accuracy, Precision, Recall, and F1 Score, enabling a comprehensive comparison to identify the most effective approach for diabetes prediction.

2 RELATED WORK

Numerous studies have explored diabetes prediction, each employing machine learning algorithms and evaluation metrics to assess model performance. For instance, Deepti Sisodia et al. [14] proposed using Naive Bayes, Decision Tree, and Naive Bayes to determine the likelihood of diabetes in patients with maximum accuracy. Accuracy, Recall, Precision, and F-Measure were utilized to evaluate the performances of the three models. The research found that Naive Bayes achieves 76.30% accuracy, the highest. Conversely, SVM and Decision Tree give an accuracy of 65.10% and 73.82%, respectively. Subhash Chandra Gupta et al. [7] employed a disease prediction model for classifying diabetic patients using machine learning classification algorithms. Decision Tree, KNN, SVM, and Random Forest algorithms were applied to create four models. The hyperparameters of the classifiers were tuned to improve their performance. The best prediction model achieved the highest F1 score of 75.68% with 88.61% accuracy. Among the four models on the dataset model D3, the Random Forest classifier performed the best. Chaitanya Sonawane et al. [17] proposed using SVM and Decision Tree to determine which approach is better for predicting diabetes. The Pima Indian diabetes dataset was used for this research. Accuracy, precision, and recall these three evaluation metrics were used to compare the performance of the models. The SVM algorithm achieved an accuracy of 76.6%, which is better than the decision tree algorithm, which achieved 75% accuracy.

Nour Abdulhadi et al. [1] utilized supervised learning methods to build a model for diabetes detection. The paper presented multiple techniques and models, including Logistic Regression, Linear Discriminant Analysis (LDA), and Linear Support Vector Machine, with accuracies ranging from 79% to 82%. The Random Forest Classifier achieved the highest accuracy of 82%. The dataset used in the research consisted of 768 instances with nine attributes, including the target variable. Priyanka Sonar and Prof. K. JayaMalini [16] proposed diabetes prediction using a machine learning approach. The paper used Decision Tree, ANN, Naive Bayes, and SVM algorithms. The models provided an accuracy of 85% for Decision Tree, 77% for Naive Bayes, and 77.3% for Support Vector Machine. Mitushi Soni and Dr. Sunita Varma [18] implemented machine learning algorithms such as K-Nearest Neighbor, Logistic Regression, Decision Tree, Support Vector Machine, Gradient Boosting and Random Forest to predict diabetes. Among all the models implemented, Random Forest showed superior performance by achieving 77% of accuracy.

A study by Amina Azrar et al. [4] explored the PIMA diabetes dataset, converting numerical data into categorical data as part of preprocessing. They applied K-Nearest Neighbors, Naïve Bayes, and Decision Tree algorithms to predict diabetes and non-diabetes instances, validating their findings through cross-validation. Their most successful result, achieving 79.56% accuracy, was obtained with the decision tree classifier using WEKA. Similarly, Aiswarya Iyer et al. [10] utilized the PIMA dataset for their predictive model, employing normalization and feature selection techniques to enhance performance. They reached their highest accuracy of 79.56% with the Naïve Bayes algorithm, implemented in the WEKA environment. Quan Zou et al. [20] also used machine learning algorithms for diabetes prediction.The dataset used in the paper contains 14 attributes which was collected from the hospital physical examination data in Luzhou, China. DecisionTree, Random Forest and Neural Network were used in the study. Five-fold cross validation was applied to evaluate the models. Random Forest achieved an accuracy of 80.84% which was the best among the other models.

3 DATASET AND ATTRIBUTES

3.1 Dataset Description

The collected dataset originated from the National Institute of Diabetes and Digestive and Kidney Diseases. The Pima Indian dataset is a renowned repository in data science and healthcare research. It consists of medical records of Pima Native Americans, particularly women aged 21 and above, residing near Phoenix, Arizona [15]. Collected by the National Institute of Diabetes and Digestive and Kidney Diseases, this dataset has been pivotal in studying diabetes onset, with variables including glucose levels, blood pressure, body mass index, and familial diabetes history. The dataset link is available here: https://www.kaggle.com/datasets/akshaydattatraykhare/diabetes-dataset

The dataset focuses on Indian women aged 21 years and older, delineating their diabetic status based on a set of defined criteria. This comprehensive dataset comprises 768 instances and encompasses 9 distinct attributes, each contributing valuable information. Of particular significance is the 'Outcome' variable, which serves as the focal point, indicating the diabetic condition of each patient. Table 1 represents the description of the dataset's attributes.

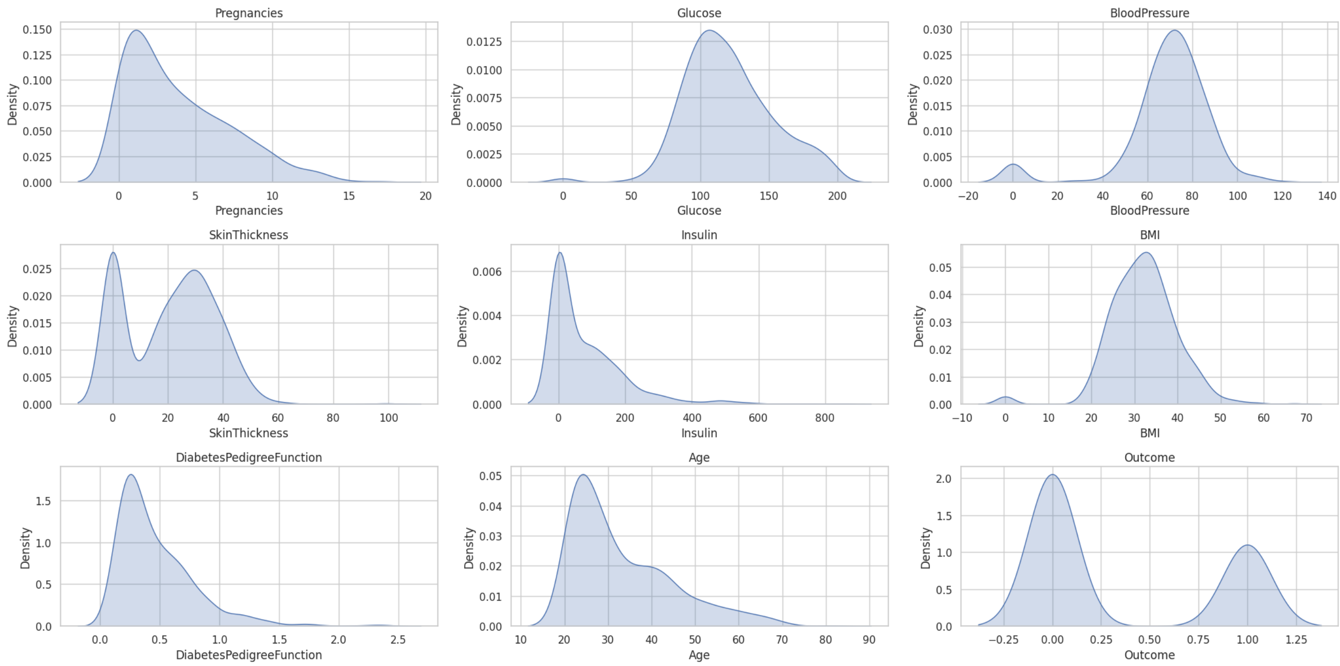

Figure 1: Density plot of the dataset

Table 1: Description of Attributes of the dataset

Attributes	Range	Description
Pregnancies	0-17	Number of pregnancies
Glucose	0-199	In an oral glucose tolerance test, plasma glucose levels after 2 hours
Blood Pressure	0-122	Diastolic blood pressure (mm Hg)
BMI	0-67.1	Body mass index = (weight in kg/(height in m)^2)
Skin Thickness	0-99	Triceps skin fold thickness (mm)
Diabetes Pedigree Function	0.078-2.42	A function that scores the likelihood of diabetes based on family history
Age	21-81	Age in years
Insulin	0-846	2-Hour serum insulin (mu U/ml)
Outcome	0-1	Class variable, diagnoses classes: 0 = healthy, 1 = diagnosed with diabetes

Density plots are invaluable tools in data visualization, particularly for understanding the distribution of continuous variables [19]. These plots provide a smooth representation of the probability density function of a given variable, offering insights into its underlying distribution characteristics. Moreover, they enable comparisons between different groups or categories within the dataset, shedding light on potential differences or similarities in their distributions.

The density graphs shown in Figure 1, offer valuable insights into diabetes detection by highlighting specific trends within the dataset. The majority of individuals have 0-5 pregnancies, with a peak around 0-1, which is relevant for understanding the impact of gestational diabetes. Glucose levels are positively skewed, peaking around 100 mg/dL, indicating most individuals are near the diabetic threshold. Blood pressure shows a normal distribution, peaking around 70-80 mmHg, relevant since high blood pressure is a diabetes risk factor. The bimodal distribution of skin thickness, peaking around 0 and 20 mm, may indicate obesity-related risks. Insulin levels are highly skewed, with most values near 0, crucial for diagnosing insulin resistance. BMI is slightly skewed with a peak around 30, highlighting obesity prevalence. The diabetes pedigree function is heavily skewed towards lower values, showing most individuals have low genetic risk. The age distribution peaks around 20-30 years, indicating younger individuals in the sample. Lastly, the bimodal outcome distribution helps understand the diabetic versus non-diabetic classification within the dataset. These patterns are essential for identifying risk factors and correlations for accurate diabetes detection.

3.2 Data Preprocessing

In this study, we implemented a series of data preprocessing methods to enhance the performance of our machine learning and deep learning models for diabetes prediction.

The initial step involved addressing missing or erroneous values within the dataset. Missing values were imputed using either the mean or median of the respective feature, depending on the distribution, to ensure no significant bias was introduced. Outliers were identified and treated to reduce their impact on model performance, ensuring a cleaner dataset for further analysis, shown in Figure 2 and Figure 3. To ensure that all features contributed equally to the model, we normalized the data. This involved scaling all numeric features to a common range, typically between 0 and 1, using Min-Max Scaling. This step was crucial for algorithms that are sensitive to the scale of data, such as Support Vector Machine (SVM) and neural networks. We derived new features from existing ones to capture more information from the data. For instance, interaction terms between features that were hypothesized to have a combined effect on the target variable were created. This step aimed to enhance the model's ability to capture complex patterns in the data, thereby improving predictive performance.

Certain features were transformed to achieve a more normal distribution, which is often assumed by many machine learning algorithms. For example, skewed features were log-transformed to reduce skewness and make the data more Gaussian-like. This transformation helped in stabilizing variance and making the patterns in the data more apparent. Given the imbalance in the dataset, with a higher number of non-diabetic cases compared to diabetic cases, techniques such as Synthetic Minority Over-sampling Technique (SMOTE) were used to balance the classes. This approach helped in preventing the model from being biased towards the majority class, ensuring better generalization. By employing these preprocessing methods, we aimed to enhance the quality of the data, reduce noise, and ensure that the features were suitable for machine learning algorithms. This comprehensive preprocessing pipeline was designed to maximize the predictive accuracy and robustness of our models in predicting diabetes.

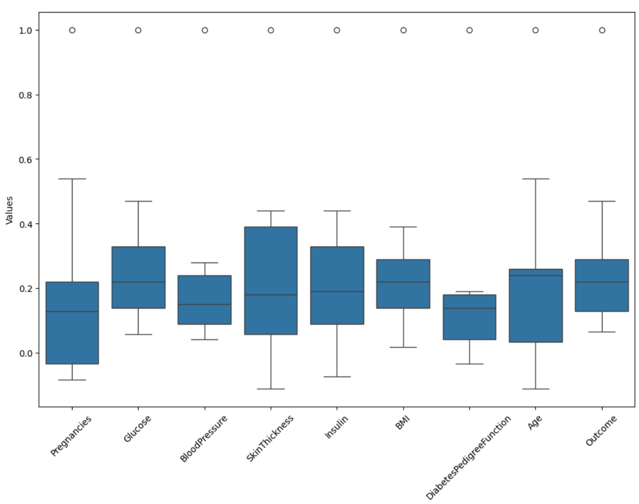

Figure 2: Box plot before preprocessing

3.3 Comparing Data Categorization Methods

Various undersampling techniques have been developed to address the challenge of class imbalance within the dataset, as illustrated in the bar graph Figure 4. The dataset in question, characterized by a significant class imbalance with 500 healthy and 268 diabetic

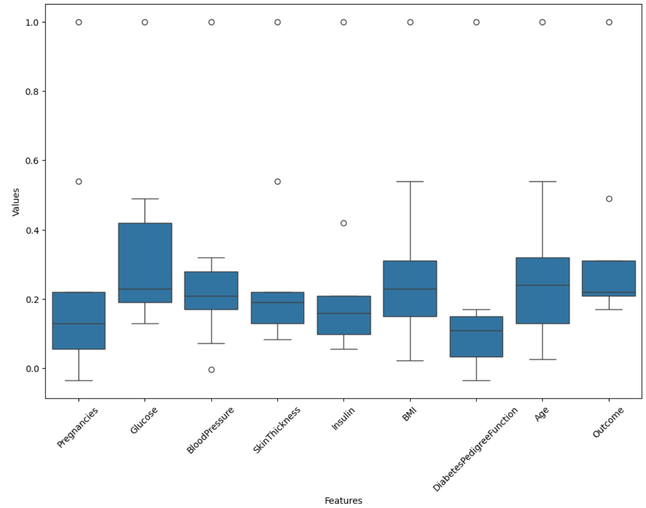

Figure 3: Box plot after preprocessing

instances, necessitated exploring undersampling strategies. This imbalance, where the number of healthy individuals considerably exceeds that of diabetic individuals, can lead to biased model performance if not adequately addressed.

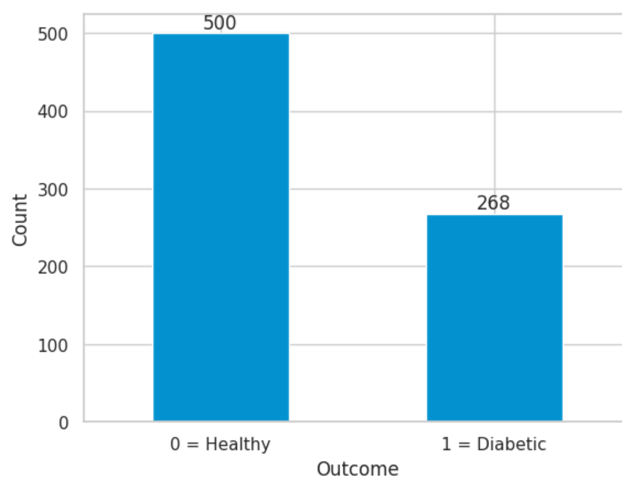

Figure 4: Distribution of Diabetes Status Categorization

Random undersampling involves the random removal of examples from the majority class (healthy) to achieve a more balanced dataset for training [9]. NearMiss-1 undersampling selects majority class samples that are closest to the minority class (diabetic), aiming to retain informative instances while reducing redundancy. Additionally, NearMiss-3 undersampling selects majority class examples that are farthest from other majority class examples, focusing on retaining diverse instances that are likely to be more informative.

These techniques help in creating a more balanced dataset, thereby improving the model's ability to accurately predict outcomes for both classes.

3.4 Data Partitioning Strategy

The dataset was split using an 80-10-10 split into training, validation, and testing sets for the purpose of experimentation. With different subsets for testing and validation to evaluate the model's generalization performance, this partitioning approach guarantees that the model is trained on the majority of data. Model training and parameter tuning were conducted using the training set, which consisted of 80% of the total data. Additionally, it allowed for robust optimization of the model's parameters and architecture. By using 10% of the data in the validation set, it was easier to pick the best model and fine-tune the hyperparameters, ensuring that the final model would perform well when applied to new data. This iterative validation process enabled selecting the best-performing model while guarding against overfitting. Finally, the testing set, also comprising 10% of the data, remained untouched during the training process and was solely used to evaluate the final model's performance on completely unseen data, providing a reliable estimate of its real-world performance.

4 Methodology

This study is structured into two distinct phases to comprehensively analyze, compare, and optimize the performance of various machine learning algorithms. In the first phase, we employ Principal Component Analysis (PCA) for feature reduction and exploration on the Decesion Tree to find the optimum number of components, followed by the implementation of three classification algorithms with the number of optimum components we found: Support Vector Machine (SVM), Random Forest, and Multilayer Perceptron Classifier (MLP).

In the second phase, we focus on optimizing the K-Nearest Neighbors (KNN) algorithm. Through systematic iteration, we determine the optimal value of K, ranging from 1 to 30, and subsequently evaluate KNN's performance using Receiver Operating Characteristic (ROC) curves.

After implementing and analyzing these models, we compare their performances to identify the best algorithm for our specific dataset. This dual-phase approach allows us to enhance the efficiency and accuracy of our predictive models, ultimately determining the most effective algorithm. The Figure 5 below represents the proposed methodology.

4.1 Feature Reduction and Algorithm Implementation

Principal Component Analysis (PCA) for Decision Tree shown in Figure 6 is a data visualization approach used to obtain deeper insights into the dataset.

The visualizations from the three figures: Figure 6, Figure 7, and Figure 8 provide critical insights into the optimal number of PCA components to be used for subsequent modeling. Figure 6, a scree plot, shows the explained variance against the number of PCA components. It illustrates that the explained variance increases significantly with the first two components and starts to level off as more components are added. This indicates that most of the variance in the data is captured by the first few components, with diminishing returns from adding more.

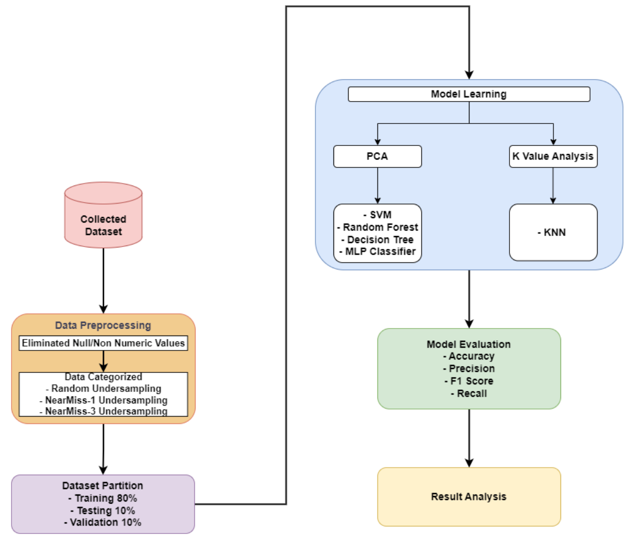

Figure 5: Workflow of the Proposed Methodology

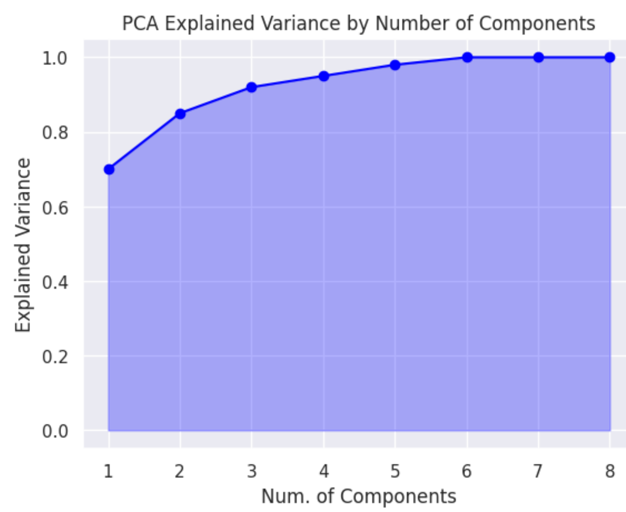

Figure 6: Variance vs. PCA Components for Decision Tree

Figure 7 depicts the error rate as a function of the number of PCA components. The error rate drops sharply when the number of components is increased from one to two, indicating a substantial improvement in model performance. After this point, the error rate decreases gradually and exhibits minor fluctuations, suggesting that adding more than six components does not significantly reduce the error rate.

Figure 8 shows the accuracy of the model as the number of PCA components varies. Similar to the error rate, the accuracy increases markedly with the first few components, peaking around six components before a slight decrease with more components. This pattern reinforces that fewer components can capture the essential variance needed for effective modeling.

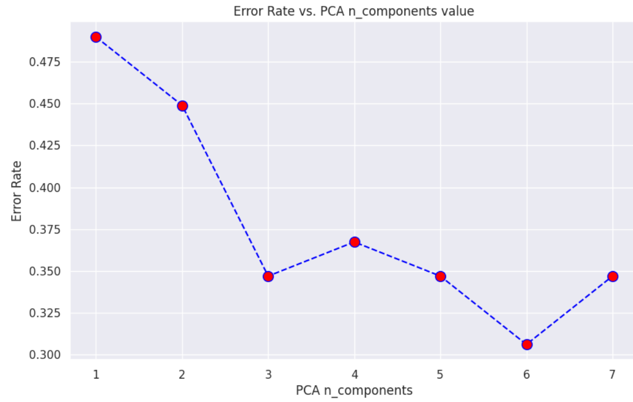

Figure 7: Error Rate vs PCA Components for Decision Tree

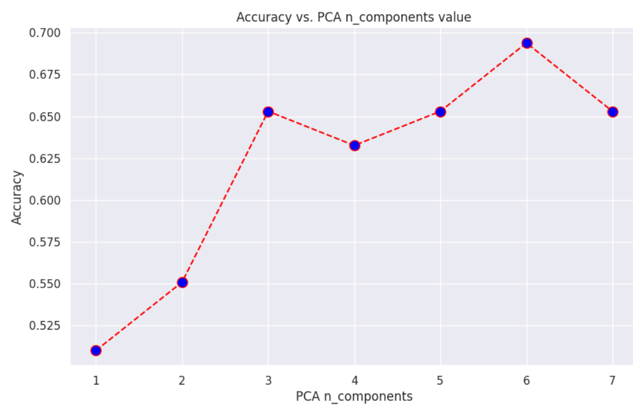

Figure 8: Accuracy vs PCA Components for Decision Tree

Six PCA components were found to have a higher explained variance with the lowest error rate and the highest accuracy shown in the figures. The scree plot (Figure 6) also shows that a further increase in components does not significantly increase explained variance. These visualizations help to comprehend the relationships among the data and the same number of components are used for each model. After this exploratory stage, three machine learning models—Decision Tree, Support Vector Machine (SVM), and Random Forest—and one deep learning model—Multilayer Perceptron Classifier (MLP)—are put into practice.

Determining the optimal number of PCA components can streamline the modeling process, improving computational efficiency without sacrificing performance. This careful selection enhances model interpretability and robustness, leading to better predictive outcomes.

4.2 Refining KNN for Improved Prediction Performance

For KNN, the K value is looped from 1 to 30 to find the best K value for the dataset.Then, the analysis found that the K-Nearest Neighbors (KNN) algorithm provided the most favorable results, as determined by the ROC curve. This suggests that, in the past investigation, KNN demonstrated the best performance in predicting diabetes based on the evaluation of the Receiver Operating Characteristic (ROC) curve.

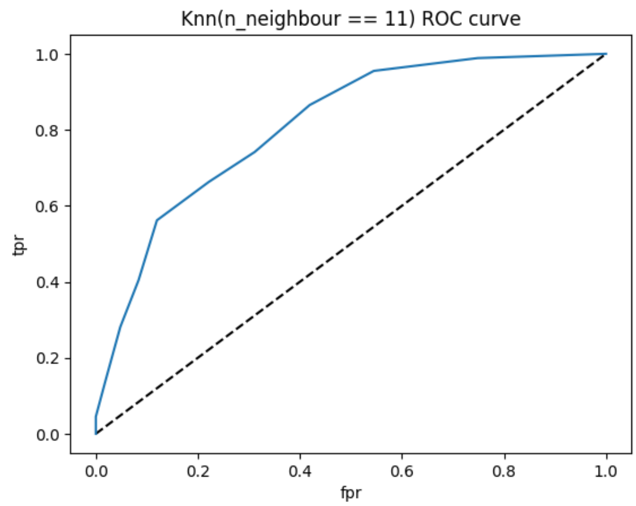

Figure 9: ROC curve for KNN

Figure 9 provides a visual representation of the Receiver Operating Characteristic (ROC) curves, revealing a consistent observation across all curves. Notably, it is evident that the K-Nearest Neighbors algorithm with 11 neighbors consistently yields the largest area under the curve. This consistent trend underscores the robustness and efficacy of the model at this specific parameter configuration, indicating superior discriminatory power in distinguishing between positive and negative instances. The discernible pattern in the ROC curves underscores the optimal performance achieved by the K-Nearest Neighbors algorithm when configured with 11 neighbors, thereby emphasizing its potential as a favorable choice for diabetes prediction within the studied dataset.

5 EXPERIMENTAL RESULTS

5.1 Performance Comparison

Following data preprocessing, five distinct models, namely Decision Tree, Support Vector Machine (SVM), Random Forest, K-Nearest Neighbor(KNN), and Multilayer Perceptron Classifier (MLP), were employed in our analysis. Each model underwent a rigorous evaluation to assess its performance and suitability for the dataset. This strategic approach allowed for a comprehensive exploration of diverse modeling techniques, enhancing the robustness of our analyses.

In the provided evaluation Table 2, various machine learning models were assessed for their performance in predicting diabetes. Among these models, the Support Vector Machine (SVM) demonstrates strong performance across both classes, with an F1-score of 0.85 for predicting diabetes and 0.76 for the absence of diabetes. The Multilayer Perceptron (MLP) also stands out, particularly in predicting the absence of diabetes, with a precision of 0.84 and an F1-score of 0.80. Additionally, the Random Forest model exhibits

Table 2: Model Performance

Model	Class	Precision	Recall	F1-score	Accuracy
SVM	No Diabetes	0.76	0.76	0.76	0.81
	Diabetes	0.85	0.85	0.85	
KNN	No Diabetes	0.79	0.88	0.83	0.77
	Diabetes	0.71	0.56	0.63	
Decison Tree	No Diabetes	0.58	0.67	0.62	0.69
	Diabetes	0.77	0.7	0.73	
Random Forest	No Diabetes	0.71	0.91	0.80	0.80
	Diabetes	0.90	0.70	0.79	
MLP	No Diabetes	0.84	0.76	0.80	0.85
	Diabetes	0.86	0.91	0.88	

competitive results, achieving an F1-score of 0.79 for predicting diabetes and 0.80 for the absence of diabetes. However, the K-Nearest Neighbors (KNN) and Decision Tree models show comparatively lower performance, with the KNN model achieving an F1-score of 0.63 for predicting diabetes and the Decision Tree model reaching an F1-score of 0.73 for the same class. Overall, upon analyzing the provided table, it is evident that the SVM model emerges as the most effective classifier for predicting diabetes.

6 Conclusion

Diabetes is one of the most common chronic illnesses in modern civilization, impacting a significant proportion of the world's population. Even if there is no 100% effective treatment, people can enjoy better lives because of early preventive and treatment methods. Deep learning and machine learning techniques have recently become more popular for diabetes prediction. Using machine learning and deep learning techniques, this work explored the field of diabetes prediction with a specific emphasis on the Pima Indian dataset. After a thorough analysis, the effectiveness of many algorithms—including Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), Random Forest, K-Nearest Neighbors (KNN), and Decision Tree—was evaluated. Evaluation criteria included essential parameters such as recall, accuracy, and precision.

Notably, the results demonstrated MLP's outstanding performance, with an accuracy rate of 85%. This demonstrated how effective MLP is in the early detection and treatment of diabetes. The study promoted ongoing research efforts to improve and optimize diabetes prediction algorithms and stressed the critical role that machine learning plays in the healthcare industry. The study's findings highlight how machine learning has the potential to revolutionize diabetes diagnosis and treatment in the future.

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ChatGPT was utilized to generate sections of this work, including text, etc.

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